

open source hardware has definitely kindled the imagination of many big players, and inspired products such as the netbook, even if the resultant products have not necessarily been open sourced: "OLPC project spawned a massive market for netbooks, which reduced the price of notebooks from Rs 40,000 to Rs 17,500, and as a spin-off brought down laptop prices by about Rs 15,000. ASUS was the first on the block with netbooks, followed by HP, LG, etc. None of these is open source hardware, but without the OLPC there would not have been netbooks."

Another example of open-source-inspired features in big brand products is the instant connect introduced in some laptops such as ASUS. This feature is brought about by an ARM SoC, not very different from the open source ECB_AT91 or Marvell ARM SoCs. It turns on and connects instantly (within two seconds) to the Net, to provide Web, e-mail and chat facilities while consuming less power than a cell-phone. These innovations were forced on the industry by the OLPC and related developments in the embedded systems market caused by FLOSH.

Open source hardware in India

Simputer remains the single 'real' example of open source hardware in India.

However, open source hardware is not otherwise unheard of in India, and it is definitely gaining traction too. Ashwin Venkat, an Indian representative of the Arduino team, recollects that an Indian has integrated Arduino with a Nokia mobile phone for data collection. "Others have used it to fly autonomous planes," he exclaims!

Open source hardware is acceptable and accessible to many design houses in India. Open specifications are available today (for example, TI has opened the specifications for OMAP3, OMAPL and other series with sample schematics and gerbers), the fundamental software to program hardware platforms is readily available, and free and open source tools and tutorials

are available, with low-cost reference designs to prototype the design.

On the flipside, Khasim explains that although there is a significant interest from Indian design houses in open source hardware, the support model for small-sector manufacturing is still inadequate. Manufacturing in India has its own problems, such as high duties and lack of government support, high cost and rarity of small-scale PCB manufacturing, shortage of ball grid array (BGA) and micro BGA assemblers, costly automated or machine-based soldering, lack of sufficient subsidies for small-scale manufacturing, incomplete ecosystem for manufacturing (everything has to be imported as there is no connector manufacturer, no inductor surface-mount device manufacturer and not even a single PCB manufacturer in India who can take care of micro vias and buried vias for a 0.4mm pitch device), and lack of a good industry body to drive 'problems to solutions' for the small-scale manufacturing sector.

Such manufacturing issues prove to be stumbling blocks in the way of small manufacturers who wish to thrive on open source hardware. Till such issues are resolved, the full potential of open source hardware cannot be tapped in India.

A promising future

"One can see the constant fragmentation in the industry, brought about by newer technologies. Consequently, markets become highly specialised, time-limited and relatively small. Conventional mass manufacturing does not work very well in such conditions. Open source hardware is a near-perfect fit in such situations. I see a very bright future for open source hardware," asserts D'Souza.

However, it is evident from Khasim's arguments that the investments in making the hardware are not always low or next to negligible like that of software. That is the reason why we do not find as many start-ups engaged in open source hardware as in the software field. But, as



Amida Simputer—an example of open source hardware

Moore's law progresses, we will see very highly integrated chips with tremendous functionality and at dirt-cheap prices, even if not free of charge.

"Hardware is atoms, software is bits. You can stop atoms at the border and bits can go right through. Hardware is not as malleable as software and the necessary smarts to make hardware, in terms of equipment, etc, are harder to come by. Having said that, a lot of the building tools are now so much commoditised that potentially in the near future, we can have really low-cost hardware—think of Fondera, Arduino, etc," says Harish Pillay, an open source evangelist based in Singapore.

Khasim sums up the Indian situation well: "The future of open source hardware is excellent, emerging, challenging. Moving forward, open source hardware will emerge as a big threat to proprietary and high-cost devices. I expect it to enter the fields of medicine, education and consumer appliances to give end-users an opportunity to afford technology at the lowest possible cost. Open source hardware, however, needs a broader and open ecosystem, individual support, training and a market to sell. As open source hardware finds its space, the manufacturing segment will also find a way to meet the time-to-market and demand-supply needs of customers, and involve itself more in research to create solutions that reach the low, mid and high segments of the society." ●

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